

Database Systems - MCQ Question Bank (Set 4)

20 Multiple Choice Questions | Includes 4 repeated from earlier sets | Answer Key on Last Page

1. A foreign key in a relational table is used to:

- A. Uniquely identify rows within the same table
- B. Enforce a link between data in two tables
- C. Speed up query execution
- D. Store encrypted data

2. Which of the following is an example of a NoSQL database?

- A. MySQL
- B. Oracle
- C. MongoDB
- D. PostgreSQL

3. Which of the following best describes a stored procedure?

- A. A precompiled set of SQL statements stored in the database and executed on demand
- B. A temporary table used in a query
- C. A type of index used for fast lookups
- D. A rule enforcing data type constraints

4. Which term describes the property that a completed transaction's changes persist even after a system failure?

- A. Atomicity
- B. Isolation
- C. Durability
- D. Consistency

5. Which of the following best describes a surrogate key?

- A. A key derived from real-world business data
- B. A system-generated unique identifier with no business meaning
- C. A key that must always be a foreign key
- D. A key that can contain duplicate values

6. Which SQL clause is used to limit the number of rows returned by a query?

- A. LIMIT
- B. TOP ONLY
- C. ROWCOUNT
- D. MAX

7. In database design, cardinality refers to:

- A. The number of columns in a table
- B. The number of instances of one entity that can relate to instances of another entity
- C. The size of the database file
- D. The number of indexes on a table

8. Which of the following best describes a materialized view?

- A. A view that is recomputed on every query

- B. A view whose result is physically stored and periodically refreshed
- C. A temporary table dropped after each session
- D. A view that cannot be indexed

9. Which lock type allows multiple transactions to read a resource but not modify it?

- A. Exclusive lock
- B. Shared lock
- C. Intent lock
- D. Update lock

10. Which of the following best describes the two-phase locking (2PL) protocol?

- A. Locks are acquired and released at random points
- B. A transaction acquires all locks first, then releases them, with no new locks after the first release
- C. Only one lock is allowed per transaction
- D. Locks are never released until the database shuts down

11. Which of the following operations is NOT part of relational algebra?

- A. Selection
- B. Projection
- C. Join
- D. Compilation

12. What is the purpose of the SQL EXPLAIN command?

- A. To encrypt query results
- B. To display the execution plan chosen by the query optimizer
- C. To grant permissions to users
- D. To create a new table

13. Which of the following best describes vertical partitioning of a table?

- A. Splitting rows across multiple tables
- B. Splitting columns of a table into separate tables
- C. Duplicating the entire table on another server
- D. Merging two tables into one

14. Which of the following is a characteristic of OLAP (Online Analytical Processing) systems?

- A. Optimized for fast, simple transactional updates
- B. Optimized for complex analytical queries over large historical data
- C. Used only for real-time order processing
- D. Does not support aggregation

15. Which of the following best describes data integrity constraints?

- A. Rules that ensure accuracy and consistency of data in a database
- B. Rules that determine physical storage location of data
- C. Rules that manage user interface design
- D. Rules that control network bandwidth

16. Which of the following best describes a covering index?

- A. An index that contains all the columns needed to satisfy a query without accessing the table

- B. An index that covers only primary key columns
- C. An index used exclusively for foreign keys
- D. An index that hides sensitive data

17. In distributed databases, the CAP theorem states that a system can only guarantee two of which three properties?

- A. Consistency, Availability, Partition tolerance
- B. Concurrency, Atomicity, Persistence
- C. Caching, Authentication, Performance
- D. Clustering, Aggregation, Parallelism

18. Which SQL statement is used to create a new table in a database?

- A. NEW TABLE
- B. CREATE TABLE
- C. MAKE TABLE
- D. ADD TABLE

19. Which of the following best describes normalization's primary trade-off?

- A. It always improves both read and write performance equally
- B. It reduces redundancy but can increase the number of joins needed for queries
- C. It eliminates the need for primary keys
- D. It removes the need for indexes

20. Which of the following best describes a database transaction log?

- A. A record of all changes made to the database, used for recovery
- B. A list of all registered users
- C. A report of query performance statistics
- D. A file storing only SELECT queries

Answer Key

1. B 2. C 3. A 4. C

5. B 6. A 7. B 8. B

9. B 10. B 11. D 12. B

13. B 14. B 15. A 16. A

17. A 18. B 19. B 20. A